# FINAL JEE–MAIN EXAMINATION – AUGUST, 2021
## Held On Wednesday 01st September, 2021
## TIME: 3:00 PM to 06:00 PM

### SECTION-A

**1.** Water sample is called cleanest on the basis of which one of the BOD values given below

(1) 11 ppm (2) 15 ppm

(3) 3 ppm (4) 21 ppm

**Official Ans. by NTA (3)**

**Sol.** Clean water could have BOD value of less than 5 ppm whereas highly polluted water could have a BOD value of 17 ppm or more.

**2.** Calamine and Malachite, respectively, are the ores of :

(1) Nickel and Aluminium

(2) Zinc and Copper

(3) Copper and Iron

(4) Aluminium and Zinc

**Official Ans. by NTA (2)**

**Sol.** Calamine $\Rightarrow ZnCO_3$

Malachite $\Rightarrow Cu(OH)_2 \cdot CuCO_3$

**3.** Experimentally reducing a functional group **cannot** be done by which one of the following reagents ?

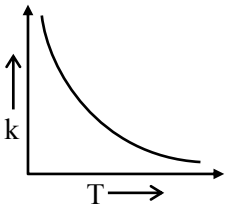
(1) Pt-C/$H_2$ (2) Na/$H_2$

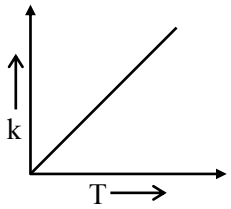
(3) Pd-C/$H_2$ (4) Zn/$H_2O$

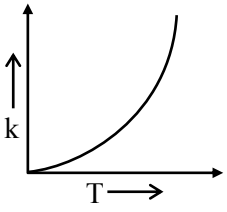
**Official Ans. by NTA (2)**

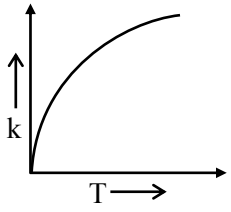
**Sol.** Solution $NaH_2$ is not reducing agent

**4.** Which one of the following given graphs represents the variation of rate constant (k) with temperature (T) for an endothermic reaction ?

(1) 


(2) 


(3) 

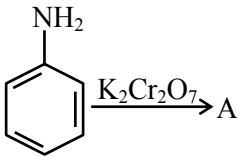

(4) 


**Official Ans. by NTA (3)**

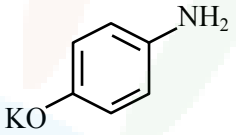
**Sol.** By observation we get this plot during measurable temperatures

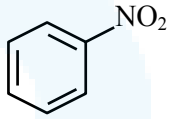
Ans. 3rd Option.

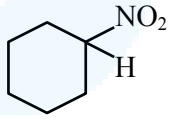
**5.** Identify A in the following reaction.



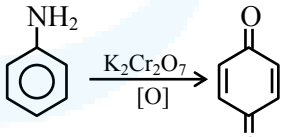

(1)


(2) 


(3) 


(4) 


**Official Ans. by NTA (1)**

**Sol.** 


Aniline [A]

**6.** In the following sequence of reactions a compound **A**, (molecular formula $C_6H_{12}O_2$) with a straight chain structure gives a $C_4$ carboxylic acid. **A** is :

$$\mathbf{A} \xrightarrow[H_3O^+]{LiAlH_4} \mathbf{B} \xrightarrow{Oxidation} C_4 - \text{carboxylic acid}$$

(1) $CH_3 - CH_2 - COO - CH_2 - CH_2 - CH_3$

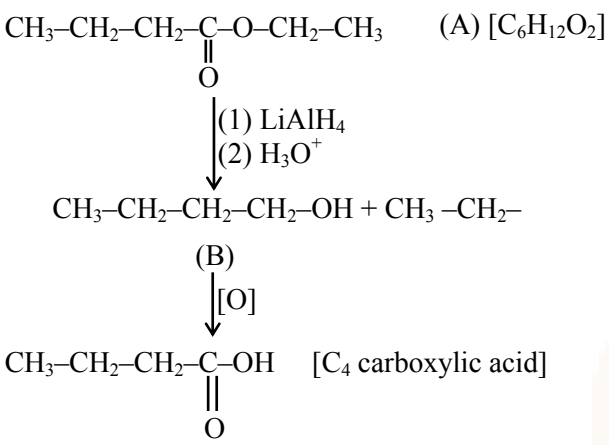
(2) $CH_3-CH_2-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_2-O-CH=CH_2$

(3) $CH_3 - CH_2 - CH_2 - COO - CH_2 - CH_3$

(4) $CH_3 - CH_2 - CH_2 - O - CH = CH - CH_2 - OH$

**Official Ans. by NTA (3)**

**Sol.**




**7.** Match **List – I** with **List - II**.

| List -I (Colloid Preparation Method) | | List -II (Chemical Reaction) | |
|---|---|---|---|
| (a) | Hydrolysis | (i) | $2AuCl_3 + 3HCHO + 3H_2O \rightarrow 2Au(sol) + 3HCOOH + 6HC1$ |
| (b) | Reduction | (ii) | $As_2O_3 + 3H_2S \rightarrow As_2S_3(sol) + 3H_2O$ |
| (c) | Oxidation | (iii) | $SO_2 + 2H_2S \rightarrow 3S(sol) + 2H_2O$ |
| (d) | Double Decomposition | (iv) | $FeCl_3 + 3H_2O \rightarrow Fe(OH)_3(sol) + 3HCl$ |

Choose the most appropriate answer from the options given below.

(1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

(2) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

(3) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

(4) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

**Official Ans. by NTA (2)**

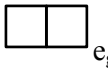
**Sol.** According to type of reactions for preparation, colloids have been classified

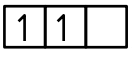
**8.** The Crystal Field Stabilization Energy (CFSE) and magnetic moment (spin-only) of an octahedral aqua complex of a metal ion ($M^{z+}$) are $-0.8\ \Delta_0$ and 3.87 BM, respectively. Identify ($M^{z+}$) :

(1) $V^{3+}$ (2) $Cr^{3+}$

(3) $Mn^{4+}$ (4) $Co^{2+}$

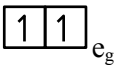
**Official Ans. by NTA (4)**

**Sol.** $V^{3+} \Rightarrow$  $_{e_g} = 2 \times 0.4\ \Delta_0$

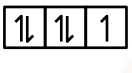
 $_{t_{2g}} = -0.8\ \Delta_0$

$= 2$ unpaired $e^-$

$\mu = 2.89$ Bm

$Co^{2+}$  $_{e_g}$ $[2 \times 0.6\ \Delta_0 - 5 \times 0.4\ \Delta_0]$

$= -0.8\ \Delta_0$

 $_{t_{2g}}$ 3 unpaired $e^- \Rightarrow \mu = 3.87$ BM

hence $d^7$ configuration is of $Co^{2+}$ Ans.

**9.** Monomer units of Dacron polymer are :

(1) ethylene glycol and phthalic acid

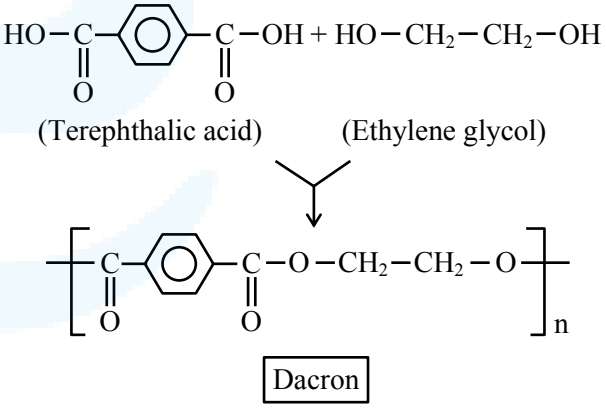
(2) ethylene glycol and terephthalic acid

(3) glycerol and terephthalic acid

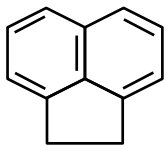
(4) glycerol and phthalic acid

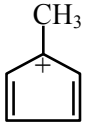
**Official Ans. by NTA (2)**

**Sol.**




**10.** Which one of the following compounds is aromatic in nature ?

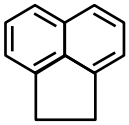
(1)  (2)

(3)  (4)

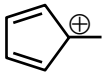
**Official Ans. by NTA (4)**

**Allen Ans. (1,4)**

**Sol.** (1) (Acenaphthene)


→ 10 $\pi e^-$ in cyclic conjugation
→ Aromatic

(2) → $4\pi e^-$ in ring conjugation ⇒ Anti Aromatic

(3)  ⇒ $4\pi e^-$ in ring conjugation ⇒ Antiaromatic

(4)

⇒ $6\pi e^-$ in ring conjugation ⇒ Aromatic

Cyclopentadienyl anion

**11.** In the given chemical reaction, colors of the $Fe^{2+}$ and $Fe^{3+}$ ions, are respectively :

$5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$

(1) Yellow, Orange (2) Yellow, Green
(3) Green, Orange (4) Green, Yellow

**Official Ans. by NTA (4)**

**Sol.** Colour of $Fe^{2+}$ is observed green and $Fe^{3+}$ is yellow

**12.** The stereoisomers that are formed by electrophilic addition of bromine to trans-but-2-ene is/are :

(1) 2 enantiomers and 2 mesomers

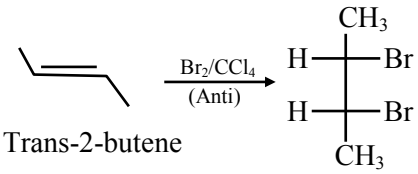
(2) 2 identical mesomers

(3) 2 enantiomers

(4) 1 racemic and 2 enantiomers

**Official Ans. by NTA (2)**

**Sol.**




Trans-2-butene

[I]

meso product

**13.** Hydrogen peroxide reacts with iodine in basic medium to give :

(1) $IO_4^-$ (2) $IO^-$ (3) $I^-$ (4) $IO_3^-$

**Official Ans. by NTA (3)**

**Sol.** $I_2 + H_2O_2 + 2OH^- \longrightarrow 2I^- + 2H_2O + O_2$

**14.** In the following sequence of reactions,

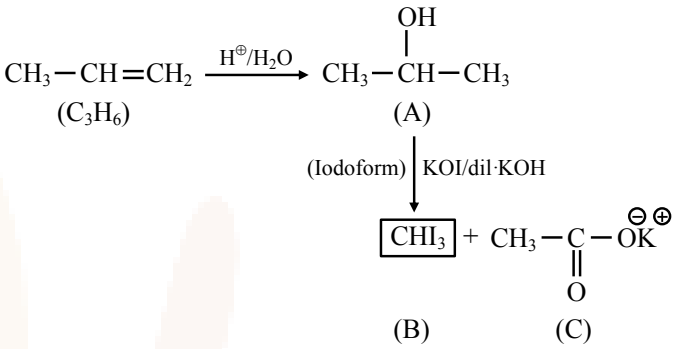
$C_3H_6 \xrightarrow{H^+/H_2O} \mathbf{A} \xrightarrow[\text{dil KOH}]{\text{KIO}} \mathbf{B} + \mathbf{C}$

The compounds **B** and **C** respectively are :

(1) $CI_3COOK$, HCOOH (2) $CI_3COOK$, $CH_3I$
(3) $CH_3I$, HCOOK (4) $CHI_3$, $CH_3COOK$

**Official Ans. by NTA (4)**

**Sol.**




**15.** Given below are **two** statements :

**Statement I :** The nucleophilic addition of sodium hydrogen sulphite to an aldehyde or a ketone involves proton transfer to form a stable ion.

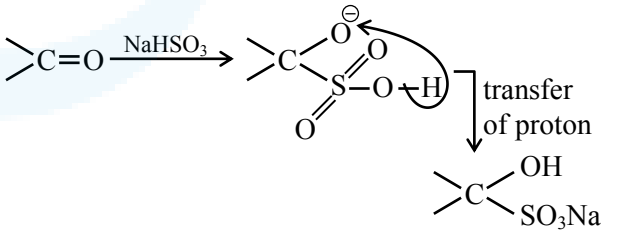
**Statement II :** The nucleophilic addition of hydrogen cyanide to an aldehyde or a ketone yields amine as final product.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **Statement I** and **Statement II** are true.
(2) **Statement I** is true but **Statement II** is false.
(3) **Statement I** is false but **Statement II** is true.
(4) Both **Statement I** and **Statement II** are false.

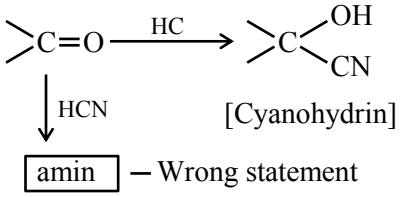
**Official Ans. by NTA (2)**

**Sol.** **Statement I :** Correct




(White crystalline soluble ppt)

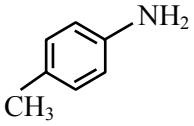
**Statement II :**

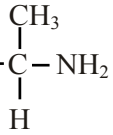
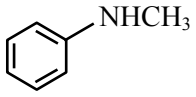



amin — Wrong statement

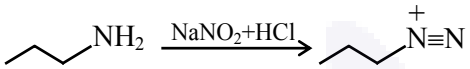
(Amine not formed)

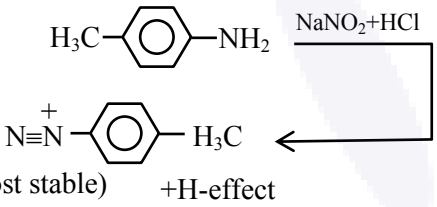
**16.** Which one of the following gives the most stable Diazonium salt ?

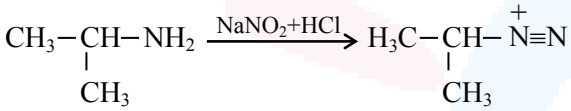
(1) $CH_3–CH_2–CH_2–NH_2$ (2) 

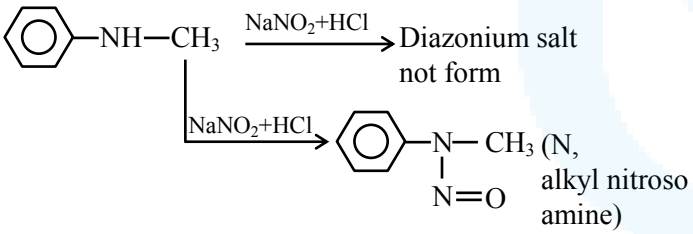
(3)  (4) 

**Official Ans. by NTA (2)**

**Sol.** (1) 

(2) 
(Most stable) +H-effect

(3) 

(4) 
Diazonium salt not form
(N, alkyl nitroso amine)

**17.** The potassium ferrocyanide solution gives a Prussian blue colour, when added to :

(1) $CoCl_3$ (2) $FeCl_2$

(3) $CoCl_2$ (4) $FeCl_3$

**Official Ans. by NTA (4)**

**Sol.** $FeCl_3 + K_4[Fe(CN)_6] \rightarrow Fe_4[Fe(CN)_6]_3$
Prussian blue

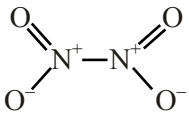
**18.** The oxide **without** nitrogen-nitrogen bond is :

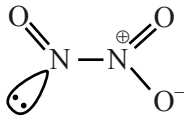
(1) $N_2O$ (2) $N_2O_4$

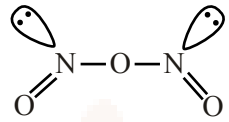
(3) $N_2O_3$ (4) $N_2O_5$

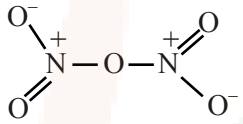
**Official Ans. by NTA (4)**

**Sol.** (1) $N\equiv N^+-O^-$

(2) 

(3)  unsymmetrical

 symmetrical

(4) 

**19.** Number of paramagnetic oxides among the following given oxides is _____.

$Li_2O$, CaO, $Na_2O_2$, $KO_2$, MgO and $K_2O$

(1) 1 (2) 2

(3) 3 (4) 0

**Official Ans. by NTA (1)**

**Sol.** $Li_2O \Rightarrow 2Li^+ \; O^{2-}$ $MgO \Rightarrow Mg^{2+} \; O^{2-}$

$CaO \Rightarrow Ca^{2+} \; O^{2-}$ $K_2O \Rightarrow 2K^+ \; O^{2-}$

$Na_2O_2 \Rightarrow 2Na^+ \; O_2^{2-}$

$KO_2 \Rightarrow K^+ \; O_2^-$

$O_2^- \Rightarrow$ Complete octet, diamagnetic

$O^{2-} \Rightarrow \sigma_{1s}^2 \; \sigma_{1s}^{*2} \; \sigma_{2s}^2 \; \sigma_{2s}^{*2} \; \sigma_{2px}^2 \; \pi_{2py}^2 \simeq \pi_{2pz}^2 \; \pi_{2py}^{*2} \simeq \pi_{2pz}^{*2}$ (dia)

$O_2^- \Rightarrow \sigma_{1s}^2 \; \sigma_{1s}^{*2} \; \sigma_{2s}^2 \; \sigma_{2s}^{*2} \; \sigma_{2px}^2 \; \pi_{2py}^2 \simeq \pi_{2pz}^2 \; \pi_{2py}^{*2} \simeq \pi_{2pz}^{*1}$ (para)

**20.** Identify the element for which electronic configuration in +3 oxidation state is $[Ar]3d^5$:

(1) Ru (2) Mn

(3) Co (4) Fe

**Official Ans. by NTA (4)**

**Sol.** $Fe^{3+}$ [Ar] $3d^5$

## SECTION-B

**1.** An empty LPG cylinder weighs 14.8 kg. When full, it weighs 29.0 kg and shows a pressure of 3.47 atm. In the course of use at ambient temperature, the mass of the cylinder is reduced to 23.0 kg. The final pressure inside of the cylinder is _____atm. (Nearest integer)

(Assume LPG of be an ideal gas)

**Official Ans. by NTA (2)**

**Sol.** Initial mass of gas = 29 –14.8 = 14.2 Kg

mass of gas used = 29 –23 = 6 Kg

gas left = 14.2 – 6 = 8.2 Kg

$$(1)\ 3.47 \times V = \left(\frac{14.2\times10^3}{M}\right) \times R \times T$$

$$(2)\ p \times V = \left(\frac{8.2\times10^3}{M}\right) \times R \times T$$

Divide :

$$\frac{(1)}{(2)} \Rightarrow \frac{3.47}{P} = \frac{14.2}{8.2}$$

P = 2.003

**2.** The molar solubility of $Zn(OH)_2$ in 0.1 M NaOH solution is $x \times 10^{-18}$ M. The value of x is _____(Nearest integer)

(Given : The solubility product of $Zn(OH)_2$ is $2 \times 10^{-20}$)

**Official Ans. by NTA (2)**

**Sol.** $Zn(OH)_2\ (s) \rightleftharpoons Zn^{+2}\ (aq) + 2OH^-\ (aq)$

S (0.1 + 2s) ≃ 0.1

$K_{sp} = S(0.1)^2$

$2 \times 10^{-20} = s \times 10^{-2} \Rightarrow s = 2 \times 10^{-18}$

$= x \times 10^{-18}$

x = 2

**3.** For the reaction $2NO_2(g) \rightleftharpoons N_2O_4(g)$, when $\Delta S = -176.0\ JK^{-1}$ and $\Delta H = -57.8\ kJ\ mol^{-1}$, the magnitude of $\Delta G$ at 298 K for the reaction is_____ $kJ\ mol^{-1}$. (Nearest integer)

**Official Ans. by NTA (5)**

**Sol.** $\Delta G = \Delta H - T\Delta S$

$$\Delta G = 57.8 - \frac{298(-176)}{1000}$$

$\Delta G = -5.352$ kJ/mole

|Nearest integer value| = 5

**4.** The sum of oxidation states of two silver ions in $[Ag(NH_3)_2]\ [Ag(CN)_2]$ complex is _____.

**Official Ans. by NTA (2)**

**Sol.** $[Ag(NH_3)_2]^+\ [Ag(CN)_2]^-$

+1 +1

**5.** The number of atoms in 8 g of sodium is $x \times 10^{23}$. The value of x is_____.(Nearest integer)

[Given : $N_A = 6.02 \times 10^{23}\ mol^{-1}$

Atomic mass of Na = 23.0 u]

**Official Ans. by NTA (2)**

**Sol.** No. of atoms = $\frac{8}{23} \times 6.02 \times 10^{23} = 2.09 \times 10^{23}$

$\simeq 2 \times 10^{23}$

$= x \times 10^{23}$

x = 2

**6.** If 80 g of copper sulphate $CuSO_4 \cdot 5H_2O$ is dissolved in deionised water to make 5 L of solution. The concentration of the copper sulphate solution is $x \times 10^{-3}$ mol $L^{-1}$. The value of x is _____.

[Atomic masses Cu : 63.54 u, S : 32 u, O : 16 u, H : 1 u]

**Official Ans. by NTA (64)**

**Sol.** Moles of $CuSO_4 \cdot 5H_2O = \frac{80}{249.54}$

$$\text{Molarity} = \frac{\frac{80}{249.54}}{5} = 64.117 \times 10^{-3}$$

Nearest integer, x = 64

**7.** A 50 watt bulb emits monochromatic red light of wavelength of 795 nm. The number of photons emitted per second by the bulb is $x \times 10^{20}$. The value of x is _____.

[Given : $h = 6.63 \times 10^{-34}$ Js and $c = 3.0 \times 10^8\ ms^{-1}$]

**Official Ans. by NTA (2)**

**Sol.** Total energy per sec. = 50 J

$$50 = \frac{n \times 6.63 \times 10^{-34} \times 3 \times 10^{8}}{795 \times 10^{-9}}$$

$n = 1998.49 \times 10^{17}$ [ n = no. of photons per second]

$= 1.998 \times 10^{20}$

$\simeq 2 \times 10^{20}$

$= x \times 10^{20}$

$x = 2$

**8.** The spin-only magnetic moment value of $B_2^+$ species is _____$\times 10^{-2}$ BM. (Nearest integer)

[Given : $\sqrt{3} = 1.73$ ]

**Official Ans. by NTA (173)**

**Sol.** $B_2^+ \Rightarrow \sigma_{1s}^2\ \sigma_{1s}^{*2}\ \sigma_{2s}^2\ \sigma_{2s}^{*2}\ \pi_{2py}^1 \simeq \pi_{2pz}^0$

$\Rightarrow 9e^-$

$\mu = \sqrt{1(1+2)} = \sqrt{3}$ BM

$= 1.73$ BM

$= 1.73 \times 10^{-2}$ BM

**9.** If the conductivity of mercury at 0°C is $1.07 \times 10^6$ S $m^{-1}$ and the resistance of a cell containing mercury is 0.243 Ω, then the cell constant of the cell is $x \times 10^4$ $m^{-1}$. The value of x is _____.(Nearest integer)

**Official Ans. by NTA (26)**

**Sol.** $k = 1.07 \times 10^6\ Sm^{-1}$, $R = 0.243\ \Omega$

$$G = \frac{1}{R} = \frac{1}{0.243}\Omega^{-1}$$

$k = G \times G^*$

$$G^* = \frac{k}{G} = \frac{1.07 \times 10^6}{\frac{1}{0.243}} \simeq 26 \times 10^4\ m^{-1}$$

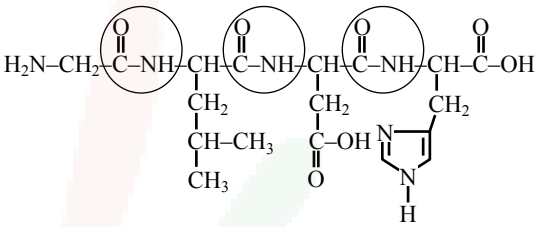
**10.** A peptide synthesized by the reactions of one molecule each of Glycine, Leucine, Aspartic acid and Histidine will have _____ peptide linkages.

**Official Ans. by NTA (3)**

**Sol.** Glycine —leucine—Aspartic acid—Histidine

peptide link  peptide link  peptide link

Total (3) peptide linkages are present



3 peptide linkage

Ans. (3)